OBJECTIVES

Project Title

Machine Learning-Based Loan Risk Prediction: A Comparative Study of Ensemble, Oversampling, and Topological Approaches for Credit and Loan Default Assessment

Prepared By

RITHISH R (23CSR178)
NAVEEN SRINIVAS V (23CSR145)
MENAHA R (23CSR130)

Department of Computer Science and Engineering

Academic Year: 2023–2027

Objectives:

The primary objective of this project is to develop an accurate, reliable, and explainable Machine Learning system to predict the loan default risk of customers. The system aims to collect loan and customer financial data from reliable sources and preprocess it by handling missing values, removing duplicate records, and transforming the data into a suitable format for analysis. Feature engineering is performed to extract important attributes such as income, credit score, debt ratio, repayment history, employment status, loan amount, and other financial indicators that significantly influence loan repayment behavior. Different Machine Learning models, including Random Forest, XGBoost, and CatBoost, are trained and compared to identify the best-performing model. The selected model is further optimized using hyperparameter tuning and data balancing techniques to improve prediction accuracy, reliability, and overall performance. The performance of each model is evaluated using standard metrics such as accuracy, precision, recall, and F1-score to ensure consistent and effective loan risk prediction.

Another major objective of this project is to make the loan prediction process transparent, efficient, and useful for both customers and financial institutions. SHAP and LIME are integrated into the system to provide clear explanations for every prediction by identifying the key factors that influence loan approval or rejection decisions. Based on the predicted risk level, the system recommends loan eligibility, a suitable loan amount, and an affordable EMI according to the customer's financial profile and repayment capacity. It also generates a customer risk score to support better credit assessment and reduce the chances of loan defaults. Finally, the complete system is implemented as a user-friendly web application that enables real-time loan risk prediction, transparent decision support, and faster loan processing, helping banks improve decision-making while providing customers with fair, reliable, and understandable loan recommendations.